

Quantized Small Judges: A Preregistered Deployment-Precision Study of a Production Verification Judge

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Abstract

Small language models are increasingly deployed as verification judges, yet their robustness to post-training quantization — the default cost-saving step in production — remains unstudied. We present a preregistered deployment-precision study of a production three-state verification judge (Qwen3-1.7B + LoRA adapter, 88.5% binary accuracy, ECE 0.072), evaluating four precision configurations $\times$ two decoding regimes over 291 held-out cases. Under a preregistered agreement gate ($\geq 99\%$ label agreement with the bf16 anchor), fp16 passes with **zero label drift** (1.0000), while int8 (0.9828, five flips) and int4-nf4 (0.9416, seventeen flips) fail the gate. Drift increases monotonically with quantization strength, corroborating an independent 3B-scale observation. We release the full per-case judgment matrix, the frozen gating criteria, and a four-rule deployment discipline for verification judges, and discuss implications for registry-based self-improving judge systems where a drifting judge silently corrupts the training signal of the entire loop.

1 Introduction

Verification judges — small models that grade the outputs of other models — are becoming infrastructure: LLM-as-judge pipelines, agent-harness evaluation services, and self-improving training loops all depend on them. Quantization is the default cost-saving step when serving such judges in production, yet the effect of post-training quantization on *judgment stability* has, to our knowledge, not been studied; a recent survey of small-language-model judges likewise notes that their robustness is under-examined.

Our perspective is narrower than “do quantized models get worse”: a judge in a registry-based self-improving loop is not a one-off evaluator but **the source of training signal** — every mis-flip it produces becomes (incorrect) training data. Quantization drift in such a judge does not merely distort one leaderboard; it silently corrupts the loop itself. This motivates treating judgment agreement, not task accuracy, as the primary metric.

We contribute: (1) the first preregistered deployment-precision verdict for a production judge, with a frozen agreement gate; (2) a four-rule deployment discipline (bf16/fp16 only for production judges, criteria-freeze, U-state honesty, two-way judging records); (3) a reproducible methodology with all per-case artifacts released.

2 Setup

- **Judge:** NACRE judge instance v1 — Qwen3-1.7B + LoRA (r16/ α 32, adapter sha16 dcbcab6f...), three-state verdicts (pass / fail / insufficient_evidence), production accuracy 88.5% (ECE 0.072).
- **Data:** 291 held-out cases, disjoint from train/dev by qid (leakage-guarded); prompts from the training template family with the judge-output channel removed (no oracle leakage).
- **Configurations:** {bf16, fp16, int8-bnb, int4-nf4} $\times$ {greedy, $T=0.3$ }; all runs on merged weights (bypassing the peft $\times$ bnb hook so the adapter is quantized with the base model, matching production serving).
- **Preregistered gate:** a precision is deployable iff greedy label agreement with bf16-greedy is $\geq 99\%$ on the frozen set (tolerance $\leq 2/291$ flips).

3 Results

Config	Three-state acc	Agreement vs bf16	Gate ($\geq 99\%$)	Verdict
bf16 (anchor)	0.8797	—	—	reference
fp16	0.8797	1.0000	pass	deployable
int8-bnb	0.8694	0.9828 (5 flips)	fail	not for prod
int4-nf4	0.8694	0.9416 (17 flips)	fail	forbidden
bf16 + $T=0.3$	—	0.9931	(disclosure only)	keep greedy

Table 1: Deployment-precision verdicts under the preregistered agreement gate.

Three observations: (1) drift grows monotonically with quantization strength; (2) an independent 3B-scale judge showed the same direction (0.8173 at 4-bit), suggesting a systematic property rather than a model-specific artifact; (3) **agreement must be measured on parsed labels, not raw output equality** — a verbatim-equality metric produces all-fail results for any quantized model and would have invalidated the study had we not caught it during script review.

4 A Deployment-Precision Discipline for Verification Judges

From these results we distill four rules, all enforced in our production deployment:

1. **Precision gate.** A judge may serve in production at a given precision iff greedy label agreement with the bf16 anchor is $\geq 99\%$ on a frozen held-out set (preregistered; may only be tightened). For our judge: fp16 yes; int8 and int4 no.
2. **Criteria freeze.** The gate, the held-out set, and the prompt template are sha-anchored before any deployment change; loosening requires re-registration.
3. **U-state honesty.** When the judge cannot decide, it must emit insufficient_evidence rather than guess; deployment changes may not trade U-rate for accuracy.

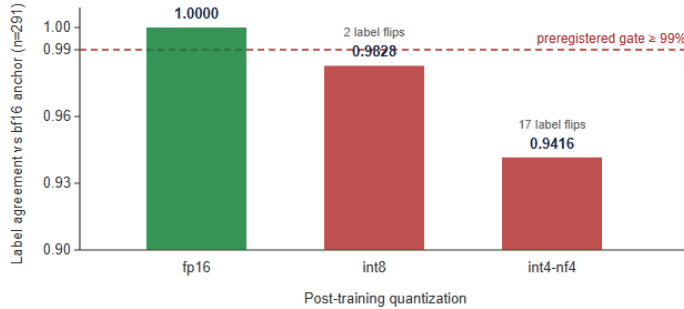


Figure 1: Label agreement vs. quantization strength.

4. **Two-way judging records.** Judges err too; every correction feeds an errata channel that retrains the judge — which is precisely why rule 1 must hold for every redeployment, or the errata loop amplifies noise instead of removing it.

Interaction with registries: judge drift is a systematic noise source in precedent registries; periodic re-anchoring on held-out sets (our L0–L3 anchor chain) bounds the drift.

5 Related Work

Each neighboring thread examines one face of the problem; none crosses them. A survey of small-language-model judges [1] flags robustness as under-studied but does not test quantization. A preregistered study of post-training quantization [2] (2026) examines welfare-relevant behaviors of general models, not judges. Rubric-preference-drift attacks (RIPD) [3] show natural-language rubrics can be manipulated — a concern orthogonal to precision, and one our criteria-freeze rule partially addresses. Harness-efficiency benchmarks such as Claw-SWE-Bench [4] measure how well agents perform under evaluation but not whether the evaluator itself survives deployment changes. Our study sits in the unoccupied intersection: quantization $\times$ small judges $\times$ deployment gating, with the loop-corruption motivation unique to registry-based systems.

References

- [1] *Small Language Models as Judges: A Survey*. OpenReview, 2026. (60+ works, SLMs $\leq 14B$; robustness noted as under-examined.)
- [2] *Does post-training quantization change welfare-relevant indicators?* — A preregistered study. LessWrong, 2026-08-10.

- [3] *Rubrics as an Attack Surface: Stealthy Preference Drift (RIPD)*, as catalogued in the promptfoo LLM Security Database.
- [4] *Claw-SWE-Bench: A Benchmark for Evaluating Coding Agent Harnesses*. arXiv:2606.12344, 2026.

6 Limitations & Negative Results

Single judge instance: generalization to other judges is claimed only as direction, not established. Soft-dimension (confidence-tier) agreement data was lost to an engineering defect in this round (non-incremental CSV write); the rerun is registered and pending. Per-case flip lists are summarized as aggregate agreement here; the full matrix is released in the artifact bundle. We also report a negative methodological result prominently rather than in a footnote: our first review of a collaborator’s replay script revealed a verbatim-equality metric that would have produced all-fail results for every quantized configuration — caught only because we re-derived the expected agreement by hand before running.

7 Artifacts

All released and sha-anchored: the frozen criteria document, the merge-then-quantize runner, the per-case judgment matrix, and run logs, in the nautilus-compass repository (<https://github.com/chunxiaox/nautilus-compass>) and the nautilus-compass Hugging Face organization (<https://huggingface.co/nautilus-compass>). The judge model card (nacre-judge-v1, adapter sha16dbcbab6fd1ff5821) carries the deployment-precision table on its face.